Algebra 1 – Module 9 Practice Test

Full Name: ______________________________
Period: _________

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Evaluate $f(x) = 5 \cdot 3^x$. Find $f(2)$.

2. Evaluate $f(x) = 24 \cdot 2^x$. Find $f(-3)$.

3. Classify each function as exponential GROWTH or DECAY, and give the percent rate of change.

a. $y = 725(1.04)^x$

b. $y = 560(0.65)^x$

4. Write an exponential function to model the situation: a town of 4,000 residents grows 6% per year.

5. A bakery sells 24 loaves of sourdough its first week, and weekly sales grow 25% each week. Predict the weekly sales after 3 weeks (round to the nearest loaf).

6. $2,500 is invested at 3% interest, compounded annually. Find the balance after 4 years (nearest cent).

7. A pond has 144 square feet of algae, and a treatment cuts the covered area in half each week. How much algae remains after 6 weeks?

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8. Is the table linear or exponential? Give the function rule.

x	y
0	7
1	28
2	112
3	448

9. Plan L starts at \$400 and adds \$90 each year. Plan E starts at \$400 and grows 35% each year. Which is worth more after 4 years? Show both values.

10. For the model $y = 480(1.22)^x$, identify the initial value and the factor, and interpret them.

11. Error Analysis — To find the value of 600 growing 15% per year for 2 years, Rex computed it by adding 30% at once: $600(1.30) = 780$. Identify his mistake and find the correct value.

12. What value does $y = 9 \cdot 2^x$ approach as x becomes very negative, and what is that line called?

13. A social media post is seen by 8 people and the number who have seen it QUADRUPLES each hour. Write an exponential model, then find how many hours until at least 512 people have seen it.

Cumulative Review — ACT Style. Circle the best answer.

14. Simplify: $\sqrt{48}$

Ⓐ $16\sqrt{3}$
Ⓑ $4\sqrt{3}$
Ⓒ $3\sqrt{8}$
Ⓓ $6\sqrt{3}$

15. Which ordered pair solves $y = x + 2$ and $x + y = 14$?

Ⓔ (8, 6)
Ⓕ (6, 8)
Ⓖ (5, 9)
Ⓗ (7, 9)

16. Solve: $3(x + 4) = 5x - 2$

Ⓐ 2
Ⓑ 7
Ⓒ 10
Ⓓ −5